

An Unsteady, Confined, Beltrami Cyclone in $\mathbb{R}^3$

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Abstract

This paper investigates the emergence and diffusion of secondary flow in an axially symmetric rotating fluid with a no-slip condition at the base and sidewall. We analyze the Beltrami field, $\vec{u}(r, z, t) = \langle -\frac{\alpha}{r}\partial_z\psi, \frac{\psi}{r}, \frac{\alpha}{r}\partial_r\psi \rangle$ as an exact solution to the Navier-Stokes equation in cylindrical coordinates with the intent to simulate the development of convection in the meridional "r - z" plane. This brief derivation of the Beltrami flow model simplifies a characteristic phenomenon known in meandering rivers, channels through a circular bend, and the "Tea-leaf" paradox explored by Einstein in 1926 [2].

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1 Introduction

Suppose that we confine a volume of initially rotating fluid within a cylindrical boundary. As the fluid rotates, it experiences shear stress against the sidewall and torsional stress at the circular base. Depending on the depth, z , the base friction can greatly impact how rapidly each velocity component decays and the degree to which convection originates. The machinery of the rotational vortex solution perceives a similar flow type to that of Bödewadt [3], though for simplicity, we will not explore the Bödewadt boundary layer.

First, we seek a general solution to the laminarized Navier-Stokes equation that satisfies these two Dirichlet boundary conditions, and second, we solve two types of initial swirls in Sections (3.1) and (3.2):

1. Irrotational, resembling that of Oseen-Lamb's vortex, and,
2. Rotational, resembling a rigid-body, centrifuged flow.

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And third, we find an approximate solution to the scalar function, $\alpha(t)$, that ensures the meridional velocities are initially zero by $\alpha(0) = 0$.

2 The Beltrami Flow Model

Consider a velocity field, $\vec{u} = \langle u_r(r, z, t), u_\theta(r, z, t), u_z(r, z, t) \rangle$ in cylindrical coordinates. Negating gravity and invoking axisymmetry ($\partial_\theta \vec{u} = 0$), the Navier-Stokes momentum equation is [10],

$$\begin{aligned} \frac{\partial u_\theta}{\partial t} + \vec{u} \cdot \nabla u_\theta &= -\frac{1}{\rho} \frac{\partial p}{\partial r} + \nu \left(\nabla^2 u_\theta - \frac{u_\theta}{r^2} \right), \quad \nabla \cdot \vec{u} = 0 \\ \equiv \frac{\partial u_\theta}{\partial t} + u_r \frac{\partial u_\theta}{\partial r} + u_z \frac{\partial u_\theta}{\partial z} + \frac{u_r u_\theta}{r} &= -\frac{1}{\rho} \frac{\partial p}{\partial r} + \nu \left(\frac{\partial^2 u_\theta}{\partial r^2} + \frac{1}{r} \frac{\partial u_\theta}{\partial r} + \frac{\partial^2 u_\theta}{\partial z^2} - \frac{u_\theta}{r^2} \right) \end{aligned} \quad (1)$$

A solution to (1) is given by a Beltrami flow, meaning the vorticity and velocity fields are parallel and proportioned by a scalar function, $\alpha(\mathbf{x}, t)$.

$$\begin{aligned} \vec{\omega} &= \nabla \times \vec{u} = \alpha \vec{u} \\ \equiv \begin{bmatrix} \omega_r \\ \omega_\theta \\ \omega_z \end{bmatrix} &= \begin{bmatrix} -\frac{\partial u_\theta}{\partial z} \\ \frac{\partial u_r}{\partial z} - \frac{\partial u_z}{\partial r} \\ \frac{1}{r} \frac{\partial(r u_\theta)}{\partial r} \end{bmatrix} = \alpha \begin{bmatrix} u_r \\ u_\theta \\ u_z \end{bmatrix} \end{aligned} \quad (2)$$

In finding a steady-state Beltrami model, Majdalani and Vyas (2003) noted that by taking the curl of the Euler equation, the flow field is independent of the pressure gradient (Equation (15) in [7]). In the unsteady case, we take the curl of Navier-Stokes,

$$\begin{aligned} \nabla \times \left[\frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \vec{u} \right] &= \nabla \times \left[-\frac{1}{\rho} \nabla p + \nu \nabla^2 \vec{u} \right] \\ \equiv \frac{\partial (\nabla \times \vec{u})}{\partial t} + \nabla \times (\vec{u} \cdot \nabla \vec{u}) &= -\underbrace{\frac{1}{\rho} \nabla \times (\nabla p)}_{=0} + \nu \nabla \times \nabla^2 \vec{u} \\ &\equiv \frac{\partial \vec{\omega}}{\partial t} + \nabla \times \left[\frac{1}{2} \nabla (\vec{u} \cdot \vec{u}) - \vec{u} \times (\nabla \times \vec{u}) \right] = \nu \nabla^2 \vec{\omega} \\ &\equiv \frac{\partial \vec{\omega}}{\partial t} + \underbrace{\frac{1}{2} \nabla \times \nabla (\vec{u} \cdot \vec{u})}_{=0} - \nabla \times (\vec{u} \times \vec{\omega}) = \nu \nabla^2 \vec{\omega} \\ &\equiv \frac{\partial \vec{\omega}}{\partial t} - \left[\underbrace{\vec{u} (\nabla \cdot \vec{\omega})}_{\nabla \cdot (\nabla \times \vec{u})=0} - \underbrace{\vec{\omega} (\nabla \cdot \vec{u})}_{=0} + (\vec{\omega} \cdot \nabla) \vec{u} - (\vec{u} \cdot \nabla) \vec{\omega} \right] = \nu \nabla^2 \vec{\omega} \\ &\equiv \frac{\partial \vec{\omega}}{\partial t} - (\vec{\omega} \cdot \nabla) \vec{u} + (\vec{u} \cdot \nabla) \vec{\omega} = \nu \nabla^2 \vec{\omega} \end{aligned}$$

and by Equation (2), $\nabla \times (\vec{u} \times \vec{\omega}) = 0$. This term likewise asserts a zero Lamb vector satisfaction, $\vec{u} \times \vec{\omega} = 0$.

As done by Majdalani and Vyas [7], one would solve the Stokes stream function, $\psi(r, z)$, in the Bragg-Hawthorne equation for a Beltrami field, $\vec{u} = \langle -\frac{1}{r} \partial_z \psi, u_\theta, \frac{1}{r} \partial_r \psi \rangle$. Although it can be useful to isolate u_θ from u_r and u_z , we seek their dependence on u_θ since the azimuthal friction with the floor changes the fluid's angular momentum, hence, generating convection [2]. Doing what is customary with the Bragg-Hawthorne equation, let $\Gamma(\psi) = r u_\theta = \psi$, and let $r u_r = -\alpha(t) \partial_z \psi$ and $r u_z = \alpha(t) \partial_r \psi$ for $\psi(r, z, t)$ (similar to Equations (23) and (24) in [10]). Let $\alpha(t)$ be the convection growth coefficient such that $\alpha(0) = 0$. Substituting each component into (1) yields the form [10],

$$\frac{\partial \psi}{\partial t} = \nu \left(\frac{\partial^2 \psi}{\partial r^2} - \frac{1}{r} \frac{\partial \psi}{\partial r} + \frac{\partial^2 \psi}{\partial z^2} \right) \quad (3)$$

which allows the cancellation of $\alpha(t)$.

2.1 Boundary Conditions:

We obtain a solution to (3) by the following Dirichlet (no-slip) boundary conditions. We can separate the variables as $\psi(r, z, t) = R(r, t)T(z, t)$, where,

$$\begin{cases} u_\theta(0, z, t) = 0, & u_\theta(R_f, z, t) = 0, \quad R_f \in \mathbb{R}^+ \\ u_\theta(r, z, 0) = \frac{1}{r}R(r, 0)T(z, 0) \neq 0, & u_\theta(r, 0, t) = 0, \quad 0 < R_f \end{cases}$$

Thus, for $R(r, t)T(z, t) \in u_\theta(r, z, t)$,

$$\begin{cases} \lim_{r \rightarrow 0} \frac{R(r, t)}{r} = 0, & T(z, 0) = \lim_{z \rightarrow \infty} T(z, t) = 1 \\ R(R_f, t) = 0, & T(0, t) = \lim_{t \rightarrow \infty} T(z, t) = 0 \\ \frac{R(r, 0)}{r} = f(r), & f(r) = u_\theta(r) \quad (= \frac{\Gamma_0}{2\pi r} \text{ or } 2\omega_0) \end{cases} \quad (4)$$

where Γ_0 is the irrotational vortex's circulation, ω_0 is the rotational vortex's angular velocity (twice its vorticity), and R_f is the sidewall radius.

3 The Stream Function Solution:

Substituting $\psi(r, z, t) = R(r, t)T(z, t)$ into Equation (3),

$$\begin{aligned} \frac{\partial (R(r, t)T(z, t))}{\partial t} &= \nu \left(\frac{\partial^2 (R(r, t)T(z, t))}{\partial r^2} - \frac{1}{r} \frac{\partial (R(r, t)T(z, t))}{\partial r} + \frac{\partial^2 (R(r, t)T(z, t))}{\partial z^2} \right) \\ \Leftrightarrow T \frac{\partial R}{\partial t} + R \frac{\partial T}{\partial t} &= \nu \left(T \frac{\partial^2 R}{\partial r^2} - \frac{T}{r} \frac{\partial R}{\partial r} + R \frac{\partial^2 T}{\partial z^2} \right) \\ \Leftrightarrow R \frac{\partial T}{\partial t} - \nu R \frac{\partial^2 T}{\partial z^2} &= \nu T \frac{\partial^2 R}{\partial r^2} - \nu \frac{T}{r} \frac{\partial R}{\partial r} - T \frac{\partial R}{\partial t} \\ \Leftrightarrow \frac{1}{T} \frac{\partial T}{\partial t} - \nu \frac{1}{T} \frac{\partial^2 T}{\partial z^2} &= \frac{\nu}{R} \frac{\partial^2 R}{\partial r^2} - \frac{\nu}{rR} \frac{\partial R}{\partial r} - \frac{1}{R} \frac{\partial R}{\partial t} \end{aligned}$$

we observe that both sides of the PDE can be separated by some constant, $k \in \mathbb{R}$. Then,

$$\begin{aligned} \frac{\partial T}{\partial t} - \nu \frac{\partial^2 T}{\partial z^2} &= kT(z, t) \\ \nu \left(\frac{\partial^2 R}{\partial r^2} - \frac{1}{r} \frac{\partial R}{\partial r} \right) - \frac{\partial R}{\partial t} &= kR(r, t) \end{aligned}$$

Each side equaling k implies that the stream function's radial diffusion is independent of its diffusion with respect to z . If $k \neq 0$ or is a function, $k(t)$, then it would be a source or sink term in both "heat" equations for $R(r, t)$ and $T(z, t)$. Because there is no source simultaneously influencing both R and T , $k = 0$, thus making R and T solutions to simpler, one-dimensional, linear diffusion equations [9]:

$$\begin{aligned} \frac{\partial T}{\partial t} &= \nu \frac{\partial^2 T}{\partial z^2} \\ \frac{\partial R}{\partial t} &= \nu \left(\frac{\partial^2 R}{\partial r^2} - \frac{1}{r} \frac{\partial R}{\partial r} \right) \end{aligned} \quad (5)$$

Solving for $T(z, t)$, suppose we use the similarity solution, $T(r, t) = f(\eta)$, where $\eta = \frac{z}{\sqrt{\nu t}}$, as derived by dimensional analysis,

$$\frac{T}{t} \sim \nu \frac{T}{z^2} \Rightarrow z \sim \sqrt{\nu t}$$

Taking each derivative of $f(\eta)$,

$$\begin{aligned}
\frac{\partial T}{\partial t} &= \frac{\partial f}{\partial \eta} \frac{\partial \eta}{\partial t} = -f' \frac{zt^{-3/2}}{2\sqrt{\nu}} = -f' \frac{\eta}{2t} \\
\frac{\partial^2 T}{\partial z^2} &= \frac{\partial}{\partial z} \left(\frac{\partial f}{\partial \eta} \frac{\partial \eta}{\partial z} \right) \\
&= \frac{\partial f}{\partial \eta} \underbrace{\frac{\partial}{\partial z} \left(\frac{\partial \eta}{\partial z} \right)}_{=0} + \frac{\partial \eta}{\partial z} \frac{\partial}{\partial z} \left(\frac{\partial f}{\partial \eta} \right) \\
&= \frac{\partial \eta}{\partial z} \cdot \frac{\partial^2 f}{\partial \eta^2} \frac{\partial \eta}{\partial z} \\
&= f'' \frac{1}{\nu t}
\end{aligned} \tag{6}$$

the PDE for $T(z, t)$ by substitution of (6) into (5) becomes,

$$\begin{aligned}
-f' \frac{\eta}{2t} &= \nu f'' \frac{1}{\nu t} \\
f'' &= -f' \frac{\eta}{2} \quad , \quad \text{let, } u(\eta) = f'(\eta) \\
\frac{du}{d\eta} &= -u \frac{\eta}{2} \\
\int \frac{du}{u} &= -\frac{1}{2} \int \eta d\eta \\
u(\eta) &= e^{-\frac{1}{4}\eta^2 + C} \\
\int df(\eta) &= C_1 \int e^{-\frac{1}{4}\eta^2} d\eta \\
f(\eta) &= C_1 \sqrt{\pi} \operatorname{erf}(\eta) + C_2 \\
T(z, t) &= C_1 \sqrt{\pi} \operatorname{erf}\left(\frac{z}{2\sqrt{\nu t}}\right) + C_2
\end{aligned}$$

Using the boundary condition (4), $\lim_{z \rightarrow \infty} T(z, t) = 1$, thus $C_1 = 1/\sqrt{\pi}$ and $C_2 = 0$. Hence, we obtain,

$$T(z, t) = \operatorname{erf}\left(\frac{z}{2\sqrt{\nu t}}\right) \tag{7}$$

Solving for $R(r, t)$, we can use separation of variables again, letting $R(r, t) = u(r)v(t)$. Our boundary conditions from (4) yield,

$$\begin{cases} R(r, 0) = u(r), & u(0) = 0 \\ R(0, t) = 0, & v(0) = 1 \end{cases}$$

Substituting this into (5), we have,

$$\begin{aligned}
\frac{d(u(r)v(t))}{dt} &= \nu \left(\frac{d^2(u(r)v(t))}{dr^2} - \frac{1}{r} \frac{d(u(r)v(t))}{dr} \right) \\
\Leftrightarrow u(r)v'(t) &= \nu \left(v(t)u''(r) - \frac{1}{r}v(t)u'(r) \right) \\
\Leftrightarrow \frac{v'(t)}{\nu v(t)} &= \frac{u''(r)}{v(r)} - \frac{u'(r)}{ru(r)}
\end{aligned}$$

We set both sides of the differential equation equal to some constant, $K \in \mathbb{R}$, for which $K \neq 0$ since we seek a nontrivial solution to R . The left-hand side becomes

$$\begin{aligned}
\frac{v'(t)}{\nu v(t)} &= K \\
\therefore v(t) &= e^{K\nu t} \quad , \quad v(0) = 1 = e^C
\end{aligned}$$

and the right-hand side is a Bessel differential equation, which has the solution in terms of first-order Bessel functions of the first and second kind.

$$\begin{aligned} K &= \frac{u''(r)}{u(r)} - \frac{u'(r)}{ru(r)} \\ \Leftrightarrow 0 &= ru''(r) - u'(r) - Kru(t) \\ \therefore u(r) &= Ar J_1(\sqrt{-K}r) + Br Y_1(\sqrt{-K}r) \end{aligned}$$

Since $\lim_{r \rightarrow 0} Y_1(\sqrt{-K}r)$ is undefined for all $K \in \mathbb{R}$, $B = 0$. Furthermore, let $K = -\bar{\lambda}^2$ if we consider only the real solutions to (3). Then $R(r, t)$ becomes a general solution,

$$R(r, t) = Cr J_1(\bar{\lambda}r) e^{-\bar{\lambda}^2 \nu t}$$

Using superposition, we can obtain all possible solutions to (3) in the form of a Fourier-Bessel series.

$$R(r, t) = \sum_{k=1}^{\infty} C_k r J_1(\bar{\lambda}_k r) e^{-\bar{\lambda}_k^2 \nu t}$$

Given the boundary condition, $u_\theta(R_f, z, t) = 0$ in (4), $\bar{\lambda}_k$ are all roots of the Bessel function such that

$$\begin{aligned} J_1(\bar{\lambda}_k r) &= 0, \quad r \in [0, 1] \\ \bar{\lambda}_k &= \frac{\lambda_k}{R_f} \Rightarrow J_1\left(\frac{\lambda_k}{R_f} r\right) = 0, \quad r \in [0, R_f] \end{aligned}$$

as computed in the Bessel function tables [1].

Table 1: Table of the first 15 roots of $J_1(x)$, where $\lambda_k \approx 3.14533k + 0.730956$.

$\lambda_1 = 3.831706$	$\lambda_6 = 19.615859$	$\lambda_{11} = 35.33231$
$\lambda_2 = 7.015587$	$\lambda_7 = 22.760084$	$\lambda_{12} = 38.47477$
$\lambda_3 = 10.173468$	$\lambda_8 = 25.903672$	$\lambda_{13} = 41.61709$
$\lambda_4 = 13.323692$	$\lambda_9 = 29.046829$	$\lambda_{14} = 44.75932$
$\lambda_5 = 16.470630$	$\lambda_{10} = 32.189680$	$\lambda_{15} = 47.90146$

Thus far, with Equation (7), the streamfunction is,

$$\psi(r, z, t) = \text{erf}\left(\frac{z}{2\sqrt{\nu t}}\right) \sum_{k=1}^{\infty} C_k r J_1\left(\frac{\lambda_k}{R_f} r\right) e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \quad (8)$$

In Sturm-Liouville theory [11], because u_θ is an eigenfunction expansion, $J_1\left(\frac{\lambda_k}{R_f} r\right)$ forms a basis of eigenfunctions in the Fourier Bessel series when $z \rightarrow \infty$ and $t \rightarrow 0$. Since in (4), $\frac{R(r, 0)}{r} = f(r)$,

$$f(r) = \sum_{k=1}^{\infty} C_k J_1\left(\frac{\lambda_k}{R_f} r\right) \quad (9)$$

we can apply the orthogonality relation [5] on a function, $f(x)$, with a domain, $x \in [0, \alpha]$. Proceeding from the orthonormal relation, the Fourier-Bessel coefficient is:

$$C_m = \frac{2}{\alpha^2 J_{\nu+1}^2(\lambda_m)} \int_0^\alpha x f(x) J_\nu\left(\frac{\lambda_m}{\alpha} x\right) dx, \quad x \in [0, \alpha] \quad (10)$$

We find these coefficients in Equation (9) by considering the two types of initial cyclonic flows: the rotational and irrotational vortices.

3.1 The Initially Rotational Vortex

For all $z > 0$, we let the constant rotational vortex be our initial condition.¹

$$u_\theta(r, 0) = f(r) = \frac{\Gamma_0}{2\pi R_f^2} r = \omega_0 r \quad (11)$$

Γ_0 is the circulation and ω_0 is the initial angular velocity of the rotational flow. Letting $f(x)$ be Equation (11), $\alpha = R_f$, and $\lambda_m = \lambda_k$, we find from using (10) to solve for C_k in (9) [4],

$$\begin{aligned} C_k &= \frac{2}{R_f^2 J_2^2(\lambda_k)} \int_0^{R_f} r (\omega_0 r) J_1\left(\frac{\lambda_k}{R_f} r\right) dr, \quad r \in [0, R_f] \\ &= \frac{2\omega_0}{R_f^2 J_2^2(\lambda_k)} \int_0^{R_f} r^2 J_1\left(\frac{\lambda_k}{R_f} r\right) dr \\ &= \frac{2\omega_0}{R_f^2 J_2^2(\lambda_k)} \frac{R_f^3}{\lambda_k^3} \int_0^{\lambda_k} u^2 J_1(u) du, \quad u = \frac{\lambda_k}{R_f} r \\ &= \frac{2\omega_0}{R_f^2 J_2^2(\lambda_k)} \frac{R_f^3}{\lambda_k^3} [u^2 J_2(u)]_0^{\lambda_k} \\ &= \frac{2\omega_0}{R_f^2 J_2^2(\lambda_k)} \frac{R_f^3}{\lambda_k^3} \lambda_k^2 J_2(\lambda_k) \\ &= \frac{2\omega_0 R_f}{\lambda_k J_2(\lambda_k)} \end{aligned}$$

the coefficients to be substituted into Equation (8). Hence, the solution is obtained.

$$u_\theta(r, z, t) = 2\omega_0 R_f \operatorname{erf}\left(\frac{z}{2\sqrt{\nu t}}\right) \sum_{k=1}^{\infty} \frac{J_1\left(\frac{\lambda_k}{R_f} r\right)}{\lambda_k J_2(\lambda_k)} e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \quad (12)$$

3.2 The Initially Irrotational Vortex

Likewise, let,

$$u_\theta(r, 0) = f(r) = \frac{\Gamma_0}{2\pi r} \quad (13)$$

be our initial condition. Using (10) again for (13), we obtain

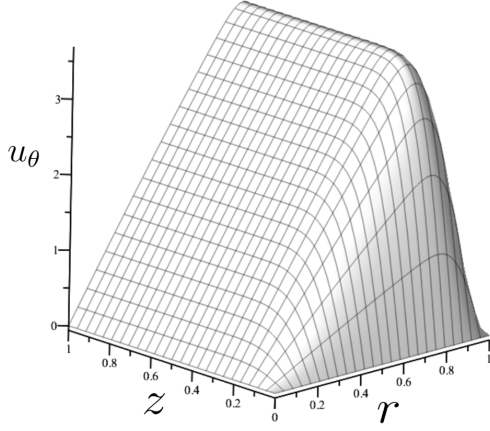
$$\begin{aligned} C_k &= \frac{2}{R_f^2 J_2^2(\lambda_k)} \int_0^{R_f} r \left(\frac{\Gamma_0}{2\pi r}\right) J_1\left(\frac{\lambda_k}{R_f} r\right) dr, \quad r \in [0, R_f] \\ &= \frac{2}{R_f^2 J_2^2(\lambda_k)} \frac{\Gamma_0}{2\pi} \int_0^{R_f} J_1\left(\frac{\lambda_k}{R_f} r\right) dr \\ &= \frac{\Gamma_0}{\pi} \frac{1}{R_f^2 J_2^2(\lambda_k)} \frac{R_f}{\lambda_k} \left[1 - J_0\left(\frac{\lambda_k}{R_f} r\right)\right]_0^{R_f} \\ &= \frac{\Gamma_0}{\pi} \frac{1 - J_0(\lambda_k)}{R_f \lambda_k J_2^2(\lambda_k)} \end{aligned}$$

the coefficients to be substituted into (8). And so,

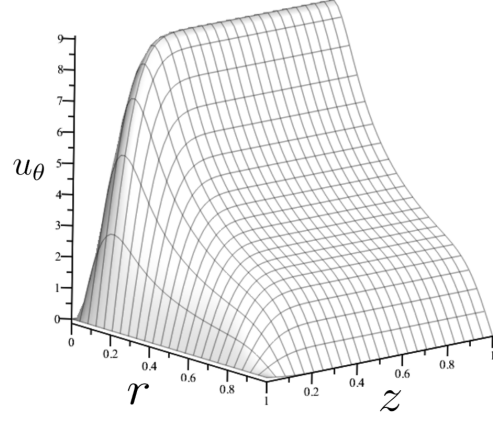
$$u_\theta(r, z, t) = \frac{\Gamma_0}{\pi} \operatorname{erf}\left(\frac{z}{2\sqrt{\nu t}}\right) \sum_{k=1}^{\infty} \frac{1 - J_0(\lambda_k)}{R_f \lambda_k J_2^2(\lambda_k)} J_1\left(\frac{\lambda_k}{R_f} r\right) e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \quad (14)$$

¹Note that $u_\theta(r, 0) = u_\theta(r, z, 0)$ since $\lim_{t \rightarrow 0} \operatorname{erf}\left(\frac{z}{2\sqrt{\nu t}}\right) = 1$, thus $\forall z > 0$, the initial condition does not vary with z .

The relation, $\frac{\Gamma_0}{2\pi R_f^2} = \omega_0$, comes from the vorticity flux, $\Gamma = \int_A \vec{\omega}_z \cdot dA = \int_0^{R_f} \omega_z \cdot 2\pi r dr \sim 2\omega_0 \pi R_f^2$, where $\|\omega_z\| \sim 2\omega_0$. In the rotational vortex, $\|\omega_z\| = 2\omega_0$, and thus, $\Gamma_0 = 2\omega_0 \pi R_f^2$.



(a) Equation (12) with $\omega_0 = 5 \text{ rad/s}$.



(b) Equation (14) with $\Gamma_0/\pi = 4 \text{ cm}^2/\text{s}$.

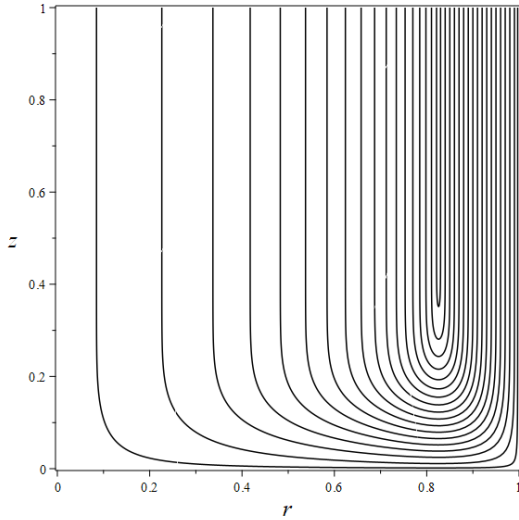
Figure 1: The initially rotational (1a) and irrotational (1b) vortices at $t = 0.5 \text{ s}$, $\nu = 0.01 \text{ cm}^2/\text{s}$, and $R_f = 1 \text{ cm}$.

4 The Meridional Flow Field

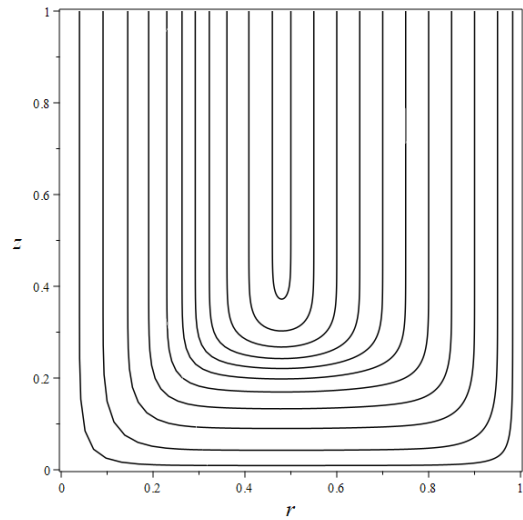
Solving $u_r = -\frac{\alpha(t)}{r} \frac{\partial \psi}{\partial z}$ and $u_z = \frac{\alpha(t)}{r} \frac{\partial \psi}{\partial r}$, we have,

$$u_r(r, z, t) = -\frac{\alpha(t)}{\sqrt{\pi \nu t}} \exp\left\{-\frac{z^2}{4\nu t}\right\} \sum_{k=1}^{\infty} C_k J_1\left(\frac{\lambda_k}{R_f} r\right) e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \quad (15)$$

$$u_z(r, z, t) = \alpha(t) \operatorname{erf}\left(\frac{z}{2\sqrt{\nu t}}\right) \sum_{k=1}^{\infty} C_k \frac{\lambda_k}{R_f} J_0\left(\frac{\lambda_k}{R_f} r\right) e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \quad (16)$$



(a) The rotational vortex with $\omega_0 = 5 \text{ rad/s}$.



(b) The irrotational vortex with $\Gamma_0/\pi = 4 \text{ cm}^2/\text{s}$.

Figure 2: Streamlines generated by Equations (15) and (16) at $t = 0.5 \text{ s}$, $\nu = 0.01 \text{ cm}^2/\text{s}$, $\alpha(t) = 1$, and $R_f = 1 \text{ cm}$. (3a) has circulation centralized in the outer region of the vortex, $r \approx 0.8$, while (3b) has circulation at $r \approx 0.5 \text{ cm}$.

4.1 Surface layer interaction at $z = h$

Suppose we seek a boundary condition on u_r and u_z at the fluid's surface, $z = h$, for some real constant, $h > 0$. In retrospect, this would ideally force $u_\theta(r, h, t) = 0$. The vertical injection-ejection meridional flow in Figure (2) is equivalent to a closed circulation due to mass conservation, and thus, there is little impact on u_θ subsisting on an azimuthal slip-condition at $z = h$. In Equation (5), decompose $T(z, t)$ into the sum two functions, $T_1(z, t)$ and $T_2(z - h, t)$, and assign the conditions,

$$\begin{cases} T(z, 0) = 1, & T(h, t) = 0 \\ T(0, t) = 0, & \lim_{t \rightarrow \infty} T(z, t) = 0 \end{cases} \quad (17)$$

Letting $T = T_1 + T_2$ describe floor and ceiling boundary layers, we have two heat equations

$$\begin{aligned} \frac{\partial T_1}{\partial t} + \frac{\partial T_2}{\partial t} &= \nu \frac{\partial^2 T_1}{\partial z^2} + \nu \frac{\partial^2 T_2}{\partial z^2} \\ \Leftrightarrow \frac{\partial T_1}{\partial t} - \nu \frac{\partial^2 T_1}{\partial z^2} &= -\frac{\partial T_2}{\partial t} + \nu \frac{\partial^2 T_2}{\partial z^2} \end{aligned}$$

with the separated solutions,

$$T_1(z, t) = \operatorname{erf}\left(\frac{z}{2\sqrt{\nu t}}\right) \quad \text{and,} \quad T_2(z - h, t) = C_1 \sqrt{\pi} \operatorname{erf}\left(\frac{z - h}{2\sqrt{\nu t}}\right) + C_2$$

Applying the boundary conditions in (17), we have,

$$T(z, t) = \operatorname{erf}\left(\frac{z}{2\sqrt{\nu t}}\right) - \left[\operatorname{erf}\left(\frac{z - h}{2\sqrt{\nu t}}\right) + \operatorname{erf}\left(\frac{h}{2\sqrt{\nu t}}\right) \right] \quad (18)$$

Applying Equation (18) to u_r and u_z , the velocity field may be written as,

$$\vec{u} = \begin{bmatrix} u_r \\ u_\theta \\ u_z \end{bmatrix} \approx \begin{bmatrix} -\frac{\alpha(t)}{\sqrt{\pi\nu t}} \left(\exp\left\{-\frac{z^2}{4\nu t}\right\} - \exp\left\{-\frac{(z-h)^2}{4\nu t}\right\} \right) \sum_{k=1}^{\infty} C_k J_1\left(\frac{\lambda_k}{R_f} r\right) e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \\ \operatorname{erf}\left(\frac{z}{2\sqrt{\nu t}}\right) \sum_{k=1}^{\infty} C_k J_1\left(\frac{\lambda_k}{R_f} r\right) e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \\ \alpha(t) \left[\operatorname{erf}\left(\frac{z}{2\sqrt{\nu t}}\right) - \operatorname{erf}\left(\frac{z-h}{2\sqrt{\nu t}}\right) - \operatorname{erf}\left(\frac{h}{2\sqrt{\nu t}}\right) \right] \sum_{k=1}^{\infty} C_k \frac{\lambda_k}{R_f} J_0\left(\frac{\lambda_k}{R_f} r\right) e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \end{bmatrix} \quad (19)$$

This constructs a symmetric pseudo-velocity field that obeys mass conservation, though it does not conclusively satisfy Equation (1) unless Equation (18) applies to u_θ in (19). The following section will consider Equations (15) and (16) in finding the secondary flow growth coefficient, $\alpha(t)$.

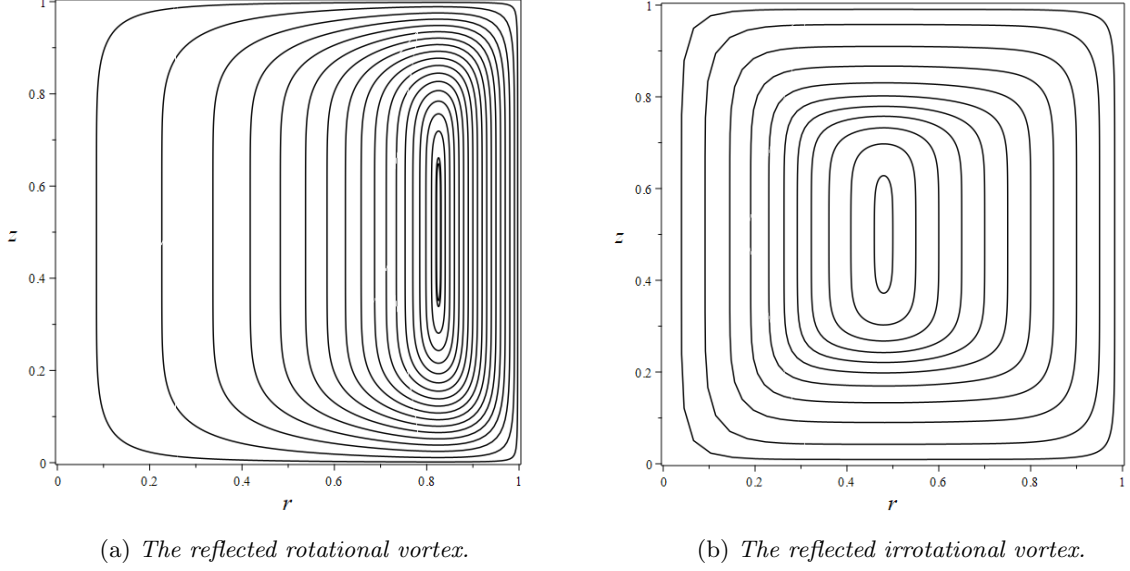


Figure 3: Streamlines generated by Equation (19). (3a) and (3b) share the same features and conditions as in Figure 2.

4.2 Finding $\alpha(t)$

Given the vector field, $\vec{u}(r, z) = \left\langle -\frac{1}{r}\partial_z\psi, \frac{\psi}{r}, \frac{1}{r}\partial_r\psi \right\rangle$, Takahashi (2023) [10] noted that if $\phi' = 0$ in,

$$\frac{\partial\psi}{\partial t} = \nu \left\{ -c^2\psi + \phi' \left(\left(\frac{\partial\psi}{\partial r} \right)^2 + \left(\frac{\partial\psi}{\partial z} \right)^2 \right) \right\} + rF_\theta$$

for $\psi(r, z)$, some function ϕ' that satisfies $\vec{\omega} \cdot \nabla\phi' = 0$, and torque F_θ , then this PDE produces,

$$\frac{\partial\alpha}{\partial t} = -\nu c^2\alpha(t)$$

and thus,

$$\alpha(t) \propto e^{-\nu c^2 t}$$

for some constant, c . Under these assumptions, this would yield an exact solution to the full Navier-Stokes equation. The diffusion, $\partial_t\alpha = -\nu c^2\alpha(t)$, can also be derived by the vorticity transport equation, $\partial_t\vec{\omega} = \nu\nabla^2\vec{\omega}$ (derived from earlier), given that $\vec{\omega} = \alpha(t)\vec{u}(r, z)$.

The difficulty that Equation (8) faces is ψ 's dependence on time; that is, substituting Equations (14), (15), and (16) into the Navier-Stokes equation. Although Equation (8) infers that

$$\alpha(t) \propto e^{-\frac{\lambda_f^2}{R_f^2}\nu t}$$

if $\vec{\omega}(r, z, t) = \alpha(t)\vec{u}(r, z)$ (Equation (2) with $\alpha \neq \text{const.}$), suppose instead that we artificially assign the boundary conditions,

$$\left\{ \begin{array}{l} \alpha(0) = 0 \\ \frac{\partial\alpha}{\partial t}\big|_{t=0} = \text{const.} \end{array} \right\} \quad (20)$$

and let $\alpha(t)$ be initially linear, such that,

$$\alpha(t) = k(t)t \quad , \quad k(t \rightarrow \infty) = 0$$

while imposing the conditions

$$|u_r| < |u_\theta| \quad \text{and,} \quad |u_z| < |u_\theta| \quad , \quad \forall r, z, t \in \mathbb{R}$$

because, for all time, the meridional velocities cannot exceed the azimuthal velocity. To ensure these conditions are satisfied, we compare the extremums of u_r and u_z to u_θ . There exists some function, $q(t) \in u_\theta$ as $z \rightarrow \infty$, such that $u_\theta(q(t), z \rightarrow \infty, t)$ is the global maximum of $u_\theta(r, z, t)$, defined as

$$u_{max}(t) := u_\theta(q(t), z \rightarrow \infty, t) = \sum_{k=1}^{\infty} C_k J_1 \left(\frac{\lambda_k}{R_f} q(t) \right) e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \quad , \quad \begin{cases} q(0) = q_0, & q_0 \in [0, R_f] \\ q(t \rightarrow \infty) = q_f, & q_0 \in (0, R_f) \end{cases}$$

For $|u_r| < u_{max}(t)$, $\exists s(t) \in u_r$ such that $u_r(s(t), 0, t)$ is the global maximum of u_r . Then,

$$\begin{aligned} |u_r(s(t), 0, t)| &< u_{max}(t) \\ \Leftrightarrow \frac{\alpha(t)}{\sqrt{\pi \nu t}} \sum_{k=1}^{\infty} C_k J_1 \left(\frac{\lambda_k}{R_f} s(t) \right) e^{-\frac{\lambda_k^2}{R_f^2} \nu t} &< \sum_{k=1}^{\infty} C_k J_1 \left(\frac{\lambda_k}{R_f} q(t) \right) e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \\ \Leftrightarrow s(t) &= q(t) \quad , \text{ thus,} \\ \therefore \alpha(t) &< \sqrt{\pi \nu t} \end{aligned}$$

which implies that $k < \sqrt{\frac{\pi \nu}{t}}$. A trivial case is letting $\alpha(t) = \sqrt{\nu t}$, however, by (17), $\alpha'(t \rightarrow 0) < \infty$.

For $|u_z| < u_{max}(t)$, the global maximum is either $u_z(0, z \rightarrow \infty, t)$ or $u_z(R_f, z \rightarrow \infty, t)$. And so,

$$\begin{aligned} &\left. \begin{aligned} &|u_z(0, z \rightarrow \infty, t)| \\ &\text{and,} \\ &|u_z(R_f, z \rightarrow \infty, t)| \end{aligned} \right\} < u_{max}(t) \\ \Leftrightarrow &\left. \begin{aligned} &\alpha(t) \sum_{k=1}^{\infty} C_k \frac{\lambda_k}{R_f} e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \\ &\text{and,} \\ &\alpha(t) \sum_{k=1}^{\infty} C_k \frac{\lambda_k}{R_f} J_0(\lambda_k) e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \end{aligned} \right\} < \sum_{k=1}^{\infty} C_k J_1 \left(\frac{\lambda_k}{R_f} q(t) \right) e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \end{aligned}$$

In the rotational vortex, $u_z(0, z \rightarrow \infty, t) < u_{max}(t)$ is trivial because $(q_0 = R_f) > 0$ and thus $J_1 \left(\frac{\lambda_k}{R_f} q_0 \right) > 0$, which is greater than $\alpha(0)$. Additionally, since $u_r(q(t), 0, t) \sim u_{max}(t)$, $u_z(0, z \rightarrow \infty, t) < u_r(q(t), 0, t)$. For brevity and the supposition of $\alpha(t)$, we will not attempt to prove all cases for u_z . Fulfilling the condition in (20), let $k = \sqrt{\frac{\pi \nu}{t + \delta}}$, such that,

$$\alpha(t) \approx \sqrt{\frac{\pi \nu t^2}{t + \delta}} \quad , \quad \begin{aligned} \delta &\in \mathbb{R}^+ \\ \delta &> 0 \end{aligned} \quad (21)$$

with,

$$\left. \frac{\partial \alpha}{\partial t} \right|_{t=0} = \lim_{t \rightarrow 0} \frac{t + 2\delta}{2\pi \nu} \left(\frac{\pi \nu}{t + \delta} \right)^{\frac{3}{2}} = \sqrt{\frac{\pi \nu}{\delta}} < \infty$$

This leaves δ as an unknown constant that governs the initial rate (and subsequently the extent) of secondary flow growth.

The true behavior of $\alpha(t)$ has yet to be explored, and will likely require numerical methods on the $\hat{r}$ and $\hat{z}$ components of Navier-Stokes. A reasonable prediction of $\alpha(t)u_{r,z}(q, s, t)$ (for brevity, $u_{r,z} = u_r$ or u_z) would take on the form in Figure (4).²

²Such an $\alpha(t)$ may also appear like $1 - e^{-c^2 \nu t}$, forming a curve, $\alpha u_{r,z} \sim (1 - e^{-c^2 \nu t}) \sum_k e^{-\frac{\lambda_k^2}{R_f^2} \nu t}$ similar to that of Figure (4).

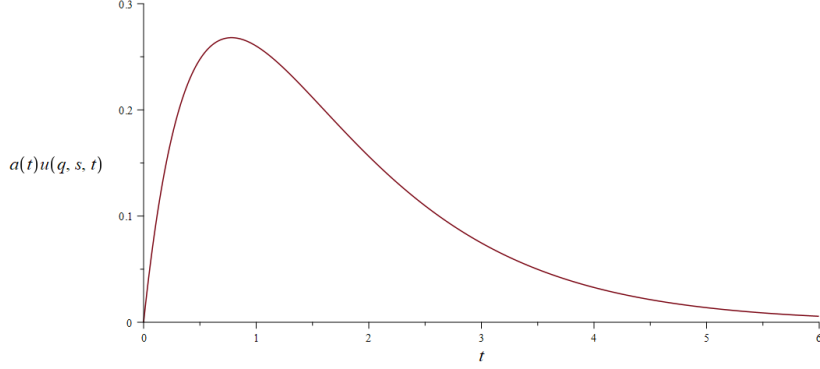


Figure 4: Equation (21) in a function, $\alpha(t)u_{r,z}(q, s, t) \sim \alpha(t) \sum_k e^{-\frac{\lambda_k^2}{R_f^2} \nu t}$, sampling at an arbitrary point, $(q, s) \in (0, R_f) \times (0, h]$, showing the growth and decay of the convection over time t .

4.2.1 Towards an exact solution:

We define a Hilbert space in the meridional domain, $\Omega = \{(r, z) : 0 < r \leq R_f, 0 \leq z \leq h, 0 \leq \theta \leq 2\pi\}$. This approach aims to perform a Galerkin Reduced Order Model (ROM) on the Navier-Stokes equations by first invoking a proper orthogonal decomposition (POD) of the form, [12]

$$\vec{u}(r, z, t) = \sum_{k=1}^n \alpha_k(t) \phi_k(r, z, t)$$

where $\phi_k = \langle \phi_{r,k}, \phi_{\theta,k}, \phi_{z,k} \rangle$ is the time-dependent basis function defined as a series of n modes in the velocity wave-function, and $\alpha_k(t)$ are the temporal coefficients to be determined. Let the k th velocities in $\vec{u}$ be,

$$\begin{aligned} u_r^{(k)}(r, z, t) &= \alpha_k(t) \phi_{r,k}(r, z, t) = -\frac{\alpha_k(t)}{\sqrt{\pi \nu t}} e^{-\frac{z^2}{4\nu t}} C_k J_1\left(\frac{\lambda_k}{R_f} r\right) e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \\ u_\theta^{(k)}(r, z, t) &= \alpha_k(t) \phi_{\theta,k}(r, z, t) = \operatorname{erf}\left(\frac{z}{2\sqrt{\nu t}}\right) C_k J_1\left(\frac{\lambda_k}{R_f} r\right) e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \\ u_z^{(k)}(r, z, t) &= \alpha_k(t) \phi_{z,k}(r, z, t) = \alpha_k(t) \operatorname{erf}\left(\frac{z}{2\sqrt{\nu t}}\right) C_k \frac{\lambda_k}{R_f} J_0\left(\frac{\lambda_k}{R_f} r\right) e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \end{aligned} \quad (22)$$

and let the POD modes be the vector, $\phi_k(r, z, t)$. If instead we had $\vec{u} = \sum_{k=1}^n \alpha_k(t) \phi_k(r, z)$ and ϕ_k forms an orthonormal basis, then, $\langle \phi_j, \phi_k \rangle = \delta_{jk}$ (where δ_{jk} is the Kronecker delta function) and α_k is solved by $\alpha_k(t) = \langle \vec{u}, \phi_k \rangle$ (see Equation 38 in [6]). The Galerkin projection of the momentum equation,

$$\frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \vec{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \vec{u} \quad (23)$$

onto a trial basis function, ϕ_j , results in the inner products,

$$\langle \phi_j, \partial_t \vec{u} \rangle + \langle \phi_j, \vec{u} \cdot \nabla \vec{u} \rangle = -\left\langle \phi_j, \frac{1}{\rho} \nabla p \right\rangle + \langle \phi_j, \nu \nabla^2 \vec{u} \rangle \quad (24)$$

where,

$$\begin{aligned}\frac{\partial \vec{u}}{\partial t} &= \sum_k \phi_k \frac{d\alpha_k}{dt} + \sum_k \alpha_k \frac{\partial \phi_k}{\partial t} \\ \nu \nabla^2 \vec{u} &= \nu \sum_k \alpha_k \nabla^2 \phi_k \\ \langle \phi_j, \vec{u} \cdot \nabla \vec{u} \rangle &= \begin{bmatrix} \langle \phi_{r,j}, u_r \frac{\partial u_r}{\partial r} + u_z \frac{\partial u_r}{\partial z} \rangle \\ \langle \phi_{\theta,j}, u_r \frac{\partial u_\theta}{\partial r} + u_z \frac{\partial u_\theta}{\partial z} + \frac{u_r u_\theta}{r} \rangle \\ \langle \phi_{z,j}, u_r \frac{\partial u_z}{\partial r} + u_z \frac{\partial u_z}{\partial z} \rangle \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ \neq 0 \\ 0 \end{bmatrix}\end{aligned}$$

The negation of the radial and zenith advection terms maintains the Beltrami condition such that u_r and u_z do not generate meridional flow irregularities. On the other hand, the azimuthal advection is satisfied by Equation (22). Because,

$$\sum_k \alpha_k \frac{\partial \phi_k}{\partial t} = \nu \sum_k \alpha_k \nabla^2 \phi_k$$

due to the viscous diffusion terms in ϕ_k , Equation (24) becomes,

$$\begin{aligned}\left\langle \phi_j, \sum_k \phi_k \frac{d\alpha_k}{dt} + \sum_k \alpha_k \frac{\partial \phi_k}{\partial t} \right\rangle &= - \left\langle \phi_j, \frac{1}{\rho} \nabla p \right\rangle + \langle \phi_j, \nu \nabla^2 \vec{u} \rangle \\ \left\langle \phi_j, \sum_k \phi_k \frac{d\alpha_k}{dt} \right\rangle + \left\langle \phi_j, \sum_k \alpha_k \frac{\partial \phi_k}{\partial t} \right\rangle &= - \left\langle \phi_j, \frac{1}{\rho} \nabla p \right\rangle + \langle \phi_j, \nu \nabla^2 \vec{u} \rangle \quad , \quad \{ \langle \mathbf{a}, \mathbf{b} + \mathbf{c} \rangle = \langle \mathbf{a}, \mathbf{b} \rangle + \langle \mathbf{a}, \mathbf{c} \rangle \} \\ \left\langle \phi_j, \sum_k \phi_k \frac{d\alpha_k}{dt} \right\rangle &= - \left\langle \phi_j, \frac{1}{\rho} \nabla p \right\rangle\end{aligned}\tag{25}$$

Taking the inner product of the pressure gradient with ϕ_j and using integration by parts, we have the volumetric integral over the region, Ω [8]:

$$\begin{aligned}\left\langle \phi_j, \frac{1}{\rho} \nabla p \right\rangle &= \frac{1}{\rho} \iiint_{\Omega} \phi_j \cdot \nabla p \, dV \\ &= \frac{1}{\rho} \iiint_{\Omega} \nabla \cdot (p \phi_j) \, dV - \frac{1}{\rho} \iint_{\Omega} p (\nabla \cdot \phi_j) \, dV \quad , \quad \{ \nabla \cdot (\mathbf{f} \mathbf{g}) = \mathbf{f} (\nabla \cdot \mathbf{g}) + \mathbf{g} \cdot (\nabla \mathbf{f}) \}\end{aligned}$$

By divergence theorem, bearing in mind that $\nabla \cdot \phi_j = 0$,

$$\iiint_{\Omega} \phi_j \cdot \nabla p \, dV = \oint_{\partial\Omega} (p \phi_j) \cdot d\mathbf{S} - \oint_{\partial\Omega} p \underbrace{(\phi_j \cdot \mathbf{n})}_{\text{div}(\phi_j)=0} \, dS = 0$$

with the second term,

$$\oint_{\partial\Omega} (p \phi_j) \cdot d\mathbf{S} \equiv \oint_{\partial\Omega} (p \phi_j) \cdot \mathbf{n} \, dS = 0$$

Therefore, the inner product projection of the pressure term is zero, leaving Equation (25) as,

$$\left\langle \phi_j, \sum_k \phi_k \frac{d\alpha_k}{dt} \right\rangle - \left\langle \phi_j, \left[\frac{u_\theta^2}{r} \right] \right\rangle = 0$$

Applying the aforementioned steps and assumptions to the $\hat{r}$ and $\hat{z}$ components of Navier-Stokes, we have,

$$\begin{aligned} \frac{\partial u_r}{\partial t} + \underbrace{u_r \frac{\partial u_r}{\partial r} + u_z \frac{\partial u_r}{\partial z}}_{=0} - \frac{u_\theta^2}{r} &= \underbrace{-\frac{1}{\rho} \frac{\partial p}{\partial r}}_{=0} + \nu \left(\nabla^2 u_r - \frac{u_r}{r^2} \right) \\ \frac{\partial u_z}{\partial t} + \underbrace{u_r \frac{\partial u_z}{\partial r} + u_z \frac{\partial u_z}{\partial z}}_{=0} &= \underbrace{-\frac{1}{\rho} \frac{\partial p}{\partial z}}_{=0} + \nu \nabla^2 u_z \\ \left(\sum_k \phi_k \frac{d\alpha_k}{dt} + \sum_k \alpha_k \frac{\partial \phi_k}{\partial t} \right) - \begin{bmatrix} \frac{u_\theta^2}{r} \\ 0 \end{bmatrix} &= \nu \begin{bmatrix} \nabla^2 u_r - \frac{u_r}{r^2} \\ \nabla^2 u_z \end{bmatrix} \\ \sum_k \phi_k \frac{d\alpha_k}{dt} &= \begin{bmatrix} \frac{u_\theta^2}{r} \\ 0 \end{bmatrix} \end{aligned}$$

Taking the Galerkin projection, we solve as follows:

$$\begin{aligned} \left\langle \phi_j, \sum_k \phi_k \frac{d\alpha_k}{dt} \right\rangle &= \left\langle \phi_j, \begin{bmatrix} \frac{u_\theta^2}{r} \\ 0 \end{bmatrix} \right\rangle \\ \int_0^{R_f} \int_0^h \phi_j \cdot \left(\sum_k \phi_k \frac{d\alpha_k}{dt} \right) r dz dr &= \int_0^{R_f} \int_0^h \phi_j^T \begin{bmatrix} \frac{u_\theta^2}{r} \\ 0 \end{bmatrix} r dz dr \\ \sum_k \frac{d\alpha_k}{dt} \int_0^{R_f} \int_0^h (\phi_j \cdot \phi_k) r dz dr &= \int_0^{R_f} \int_0^h (\phi_{r,j} u_\theta^2 + 0) dz dr \\ \sum_k \dot{\alpha}_k(t) \langle \phi_j, \phi_k \rangle &= \int_0^{R_f} \int_0^h \phi_{r,j}(r, z, t) u_\theta(r, z, t)^2 dz dr \\ \sum_k \dot{\alpha}_k(t) \mathcal{M}_{jk}(t) &= \mathcal{F}_j(t) \end{aligned}$$

Defining,

$$\begin{aligned} u_\theta(r, z, t)^2 &:= \text{erf} \left(\frac{z}{2\sqrt{\nu t}} \right)^2 \left[\sum_k C_k J_1 \left(\frac{\lambda_k}{R_f} r \right) e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \right]^2 \\ \mathcal{M}_{jk}(t) &:= \langle \phi_j, \phi_k \rangle = \iint_\Omega (\phi_j \cdot \phi_k) dV = \int_0^{R_f} \int_0^h (\phi_{r,j} \phi_{r,k} + \phi_{z,j} \phi_{z,k}) r dz dr \\ \mathcal{F}_j(t) &:= \left\langle \phi_j, \begin{bmatrix} \frac{u_\theta^2}{r} \\ 0 \end{bmatrix} \right\rangle = \int_0^{R_f} \int_0^h \phi_{r,j}(r, z, t) u_\theta(r, z, t)^2 dz dr \end{aligned}$$

denoting $\mathcal{F}$ as the centripetal force term. If we consider the k th velocity term, $u_\theta^{(k)}$, in $\mathcal{M}_{kk}$ and $\mathcal{F}_k$, a k th growth coefficient, α_k , emerges as,

$$\alpha_k(t) = \int_0^t \frac{\mathcal{F}_k(t)}{\mathcal{M}_{kk}(t)} dt \quad (26)$$

Solving for $\mathcal{F}_k(t)$,

$$\begin{aligned} \mathcal{F}_k(t) &= \int_0^h \int_0^{R_f} \phi_{r,k} \left(u_\theta^{(k)} \right)^2 dr dz \\ &= \int_0^h \int_0^{R_f} \frac{1}{\sqrt{\pi \nu t}} e^{-\frac{z^2}{4\nu t}} C_k J_1 \left(\frac{\lambda_k}{R_f} r \right) e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \text{erf}^2 \left(\frac{z}{2\sqrt{\nu t}} \right) C_k^2 J_1^2 \left(\frac{\lambda_k}{R_f} r \right) e^{-2\frac{\lambda_k^2}{R_f^2} \nu t} dr dz \\ &= \frac{C_k^3}{\sqrt{\pi \nu t}} e^{-3\frac{\lambda_k^2}{R_f^2} \nu t} \int_0^h e^{-\frac{z^2}{4\nu t}} \text{erf}^2 \left(\frac{z}{2\sqrt{\nu t}} \right) dz \int_0^{R_f} J_1^3 \left(\frac{\lambda_k}{R_f} r \right) dr \\ &= C_k^3 e^{-3\frac{\lambda_k^2}{R_f^2} \nu t} \frac{1}{3} \text{erf}^3 \left(\frac{h}{2\sqrt{\nu t}} \right) \int_0^{R_f} J_1^3 \left(\frac{\lambda_k}{R_f} r \right) dr \end{aligned}$$

In a large R_f regime, we have the Bessel function integral, $\int_0^\infty J_1(x)dx = \frac{\sqrt{3}}{2\pi}$ (pg. 389 of [4]). Thus,

$$\mathcal{F}_k(t) \approx C_k^3 e^{-3\frac{\lambda_k^2}{R_f^2}\nu t} \frac{1}{3} \frac{\sqrt{3}}{2\pi} \operatorname{erf}^3\left(\frac{h}{2\sqrt{\nu t}}\right)$$

Likewise, we can solve $\mathcal{M}_{kk}(t)$ as follows:

$$\begin{aligned} \mathcal{M}_{kk}(t) &= \int_0^{R_f} \int_0^h (\phi_{r,k}^2 + \phi_{z,k}^2) r dz dr \\ &= \int_0^{R_f} \int_0^h \left(\frac{1}{\sqrt{\pi\nu t}} e^{-\frac{z^2}{4\nu t}} C_k J_1\left(\frac{\lambda_k}{R_f} r\right) e^{-\frac{\lambda_k^2}{R_f^2}\nu t} \right)^2 r dz dr + \\ &\quad \int_0^{R_f} \int_0^h \left(\operatorname{erf}\left(\frac{z}{2\sqrt{\nu t}}\right) C_k \frac{\lambda_k}{R_f} J_0\left(\frac{\lambda_k}{R_f} r\right) e^{-\frac{\lambda_k^2}{R_f^2}\nu t} \right)^2 r dz dr \\ &= C_k^2 \frac{1}{\pi\nu t} e^{-2\frac{\lambda_k^2}{R_f^2}\nu t} \int_0^h e^{-2\frac{z^2}{4\nu t}} dz \int_0^{R_f} J_1^2\left(\frac{\lambda_k}{R_f} r\right) r dr + \\ &\quad C_k^2 \frac{\lambda_k^2}{R_f^2} e^{-2\frac{\lambda_k^2}{R_f^2}\nu t} \int_0^h \operatorname{erf}^2\left(\frac{z}{2\sqrt{\nu t}}\right) dz \int_0^{R_f} J_0^2\left(\frac{\lambda_k}{R_f} r\right) r dr \\ &= C_k^2 \frac{1}{\pi\nu t} e^{-2\frac{\lambda_k^2}{R_f^2}\nu t} \frac{1}{\sqrt{2\pi\nu t}} \operatorname{erf}\left(\frac{h}{\sqrt{2\nu t}}\right) \frac{R_f^2}{2} J_0^2(\lambda_k) + \\ &\quad C_k^2 \frac{\lambda_k^2}{R_f^2} e^{-2\frac{\lambda_k^2}{R_f^2}\nu t} \frac{R_f^2}{2} J_0^2(\lambda_k) \int_0^h \operatorname{erf}^2\left(\frac{z}{2\sqrt{\nu t}}\right) dz, \quad (\text{see pg. 270 of [4]}) \\ &= C_k^2 e^{-2\frac{\lambda_k^2}{R_f^2}\nu t} \frac{R_f^2}{2\pi\nu t \sqrt{2\pi\nu t}} \operatorname{erf}\left(\frac{h}{\sqrt{2\nu t}}\right) J_0^2(\lambda_k) + \\ &\quad C_k^2 \frac{\lambda_k^2}{2} e^{-2\frac{\lambda_k^2}{R_f^2}\nu t} J_0^2(\lambda_k) \int_0^h \operatorname{erf}^2\left(\frac{z}{2\sqrt{\nu t}}\right) dz \end{aligned}$$

with,

$$\begin{aligned} \int_0^h \operatorname{erf}^2\left(\frac{z}{2\sqrt{\nu t}}\right) dz &= 4\sqrt{\frac{\nu t}{\pi}} e^{-\frac{h^2}{4\nu t}} \operatorname{erf}\left(\frac{h}{2\sqrt{\nu t}}\right) + h \operatorname{erf}^2\left(\frac{h}{2\sqrt{\nu t}}\right) - 2\sqrt{\frac{2\nu t}{\pi}} \operatorname{erf}\left(\frac{h}{\sqrt{2\nu t}}\right) \\ &\approx h - 2\sqrt{\frac{2\nu t}{\pi}} \end{aligned}$$

if h is relatively large. Concisely and approximately,

$$\begin{aligned} \mathcal{M}_{kk}(t) &\approx C_k^2 e^{-2\frac{\lambda_k^2}{R_f^2}\nu t} J_0^2(\lambda_k) \frac{R_f^2}{2\pi\nu t \sqrt{2\pi\nu t}} \operatorname{erf}\left(\frac{h}{\sqrt{2\nu t}}\right) + C_k^2 \frac{\lambda_k^2}{2} e^{-2\frac{\lambda_k^2}{R_f^2}\nu t} J_0^2(\lambda_k) \left(h - 2\sqrt{\frac{2\nu t}{\pi}} \right) \\ &\approx C_k^2 e^{-2\frac{\lambda_k^2}{R_f^2}\nu t} J_0^2(\lambda_k) \left[\frac{R_f^2}{2\pi\nu t \sqrt{2\pi\nu t}} \operatorname{erf}\left(\frac{h}{\sqrt{2\nu t}}\right) + h - 2\sqrt{\frac{2\nu t}{\pi}} \right] \end{aligned}$$

Substituting in these approximations into Equation (26),

$$\begin{aligned} \alpha_k(t) &\approx \int_0^t \frac{C_k^3 e^{-3\frac{\lambda_k^2}{R_f^2}\nu t} \frac{1}{3} \frac{\sqrt{3}}{2\pi} \operatorname{erf}^3\left(\frac{h}{2\sqrt{\nu t}}\right)}{C_k^2 e^{-2\frac{\lambda_k^2}{R_f^2}\nu t} J_0^2(\lambda_k) \left[\frac{R_f^2}{2\pi\nu t \sqrt{2\pi\nu t}} \operatorname{erf}\left(\frac{h}{\sqrt{2\nu t}}\right) + h - 2\sqrt{\frac{2\nu t}{\pi}} \right]} dt \\ &\approx \int_0^t \frac{C_k e^{-\frac{\lambda_k^2}{R_f^2}\nu t} \frac{1}{3} \frac{\sqrt{3}}{2\pi} \operatorname{erf}^3\left(\frac{h}{2\sqrt{\nu t}}\right)}{J_0^2(\lambda_k) \left[\frac{R_f^2}{2\pi\nu t \sqrt{2\pi\nu t}} \operatorname{erf}\left(\frac{h}{\sqrt{2\nu t}}\right) + h - 2\sqrt{\frac{2\nu t}{\pi}} \right]} dt \end{aligned}$$

and given that h is large, $\mathcal{F}_k$ governs over $\mathcal{M}_{kk}$ in large time. Let $\mathcal{M}_{kk}(t) = M_{kk} \in \mathbb{R}$, and $h \in \mathcal{F}_k$ also be large. Given the boundary condition, $\alpha_k(0) = 0$, the integrand roughly becomes,

$$\begin{aligned}\alpha_k(t) &\approx \frac{1}{M_{kk}} \int_0^t C_k e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \frac{1}{3} \frac{\sqrt{3}}{2\pi} dt \\ &\approx \frac{1}{M_{kk}} C_k \frac{1}{3} \frac{\sqrt{3}}{2\pi} \int_0^t e^{-\frac{\lambda_k^2}{R_f^2} \nu t} dt \\ &\approx \frac{1}{M_{kk}} C_k \frac{1}{3} \frac{\sqrt{3}}{2\pi} \frac{R_f^2}{\lambda_k^2 \nu} \left(1 - e^{-\frac{\lambda_k^2}{R_f^2} \nu t}\right)\end{aligned}$$

with a closed-form pseudo-solution,

$$\alpha_k(t) \approx \frac{C_k R_f^2 \sqrt{3}}{6\pi M_{kk} \lambda_k^2 \nu} \left(1 - e^{-\frac{\lambda_k^2}{R_f^2} \nu t}\right) = U_k \left(1 - e^{-\frac{\lambda_k^2}{R_f^2} \nu t}\right) \quad (27)$$

letting $U_k \in \mathbb{R}^+$, $0 < U_k < 1$, be a velocity constant that does not interfere with the C_k present in $u_r^{(k)}$ and $u_z^{(k)}$. While operating under loose assumptions, Equation (27) affirms the prediction in (20): the finite initial growth rate. This approximation also captures the convergence of α_k as $t \rightarrow \infty$; a more realistic feature that is absent in the previous attempt, (21).

5 Conclusions

While this Beltrami flow model simplifies a complex boundary interaction phenomenon in swirl dynamics, it operates under numerous assumptions that would be difficult to idealize in CFD simulations, such as the natural emergence of boundary-layer separation at the cylindrical base and its instability in large time. A few of these assumptions include,

1. The Lamb vector is always zero, such that $\vec{\omega} \times \vec{u} = 0$ (defining Equation (2)).
2. The meridional flow is vertically symmetric about the r -plane, on which Equation (19) relies.
3. u_r and u_z can freely slip at boundaries u_θ does not, that being $u_r(r, 0, t) \neq 0$ and $u_z(R_f, z, t) \neq 0$.
4. $\alpha(t)$ is initially linear, is bounded below $\sqrt{\pi \nu t}$, and is largely assumed to work under the condition, (20). For brevity, the behavior of the exact $\alpha_k(t)$ is not explored in this paper.
5. The gravitational force is negated, which may influence the meridional flow geometry significantly. For example, the cyclone's updraft (generated by u_z) may significantly decrease with respect to increasing z under gravitational resistance.

References

- [1] C. L. BEATTE, *Table of first 700 zeros of bessel functions $j(x)$ and $j'(x)$* , (August 20, 1957), pp. 691–697.
- [2] A. EINSTEIN, *The cause of the formation of meanders in the courses of rivers and of the so-called baer's law*, Prussian Academy, 14 (January 7, 1926), p. 106.
- [3] E. A. EMILIA CRESPO DEL ARCO, *Stability, transition and turbulence in rotating cavities*, WIT Press, 41 (January, 2004), p. 2.
- [4] W. R. ERNST AND H. JENA, *Tables of some indefinite integrals of bessel functions of integer order*, (2019), p. 15.
- [5] R. L. HERMAN, *Introduction to Partial Differential Equations, 5.5: Fourier-Bessel Series*, LibreTexts, 2025.

- [6] E. A. KUNIIHIKO TAIRA, *Modal analysis of fluid flows: An overview*, AIAA, (October 31, 2017).
- [7] J. MAJDALANI AND A. VYAS, *Exact solution of the bidirectional vortex*, AIAA Journal, 44 (October, 2006), pp. 2210–2211.
- [8] C. A. MANOGUE AND T. DRAY, *The Geometry of Vector Calculus*, Oregon State University (OSU), 2009.
- [9] A. D. POLYANIN, *Handbook of linear partial differential equations for engineers and scientists*, CRC Press LLC, (2002), pp. 76–77.
- [10] K. TAKAHASHI, *Three-dimensional unsteady axisymmetric viscous beltrami vortex solutions to the navier–stokes equations*, MDPI, (May 22, 2023), pp. 460–476.
- [11] W. F. TRENCH, *Elementary Differential Equations with Boundary Values Problems, 13.2: Sturm-Liouville Problems*, LibreTexts, 2024.
- [12] E. A. VILAS SHINDE, *A galerkin-free model reduction approach for the navier–stokes equations*, HAL Open Science, (January 11, 2021).